Sensor networks

UAV-Empowered IoT Network With Hardware Impairments and Shadowing

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Abstract—This letter investigates the performance of an unmanned aerial vehicle (UAV)-enabled communication network for energy-constrained Internet of Things devices (IoDs) in the presence of transceiver hardware impairments. Herein, a UAV operates as a radio frequency energy harvesting transmitter for several IoDs and employs the decode-and-forward protocol to relay the gathered information from the IoDs to the base station. We employ a first-order autoregressive method for modeling the relative mobility between the nodes. The log-normal distribution is used to characterize the shadowing components for modeling the composite fading channel. We leverage the Gauss–Hermite quadrature method to obtain accurate expressions for the outage probability, followed by system throughput and energy efficiency.

Index Terms—Sensor networks, hardware impairments, Internet of Things (IoT) devices (IoDs), outage probability, unmanned aerial vehicle (UAV) relaying, wireless power transfer (WPT).

I. INTRODUCTION

Internet of Things (IoT) technology is evolving expeditiously with the inclusion of compact, intelligent, and widely distributed sensor devices that are being utilized to bridge the physical and virtual worlds [1]. These devices are often used in hard-to-reach locations, such as territorial borders, remote locations with poor connectivity and powering infrastructure, emergencies, and natural or man-made calamities. In such crucial and adverse circumstances, gathconcering and disseminating the data are challenging tasks. Furthermore, the performance of these battery-driven IoT sensors is constrained by their energy budget [2]. Recently, energy harvesting (EH) from radio-frequency (RF) has been widely employed to enhance the wireless network's lifetime, as the other traditional EH approaches are largely affected by atmospheric and environmental conditions. To this end, power splitting and time-switching (TS) receiver architectures are being extensively used and studied in literature for accomplishing simultaneous wireless information and power transfer (SWIPT). On the other hand, recent research has shown that the system performance can be severely affected by a variety of hardware imperfections that are associated with the RF transceivers [3], [4]. Thus, taking these hardware impairments (HIs) into account makes a more realistic approach in assessing the system performance accurately.

On another front, unmanned areal vehicles (UAVs) as airborne nodes offer great potential to establish on-demand communication links in natural calamities, search and rescue operations, and various civilian and military applications [5]. The UAVs can perform multiple tasks, e.g., they can serve as wireless power transmitter to extend the lifespan of IoT devices (IoDs). In addition, they can operate as an aerial node for gathering information from energy-constrained IoDs and relaying it to the network [6]. Here, the low-power data transmission from the multiple IoDs located in a small-sized cluster can be made efficient by using proper selection techniques, such as based on best signal-to-noise ratio (SNR) [7]. In recent years, UAV-aided communication networks have received extensive interest in accomplishing many operations, such as enabling wireless power transfer (WPT) and relaying assistance. Specifically, the authors in [8] considered the effects of HIs on the performance of the two-way relaying network over the Nakagami-m fading channel. Further, Indu et al. [9] studied the path-loss scenarios for the UAVs and offered a projectile motion-based trajectory to monitor the agricultural process. Likewise, the system performance analysis with the probabilistic path-loss model has been carried out in [10].

Motivated by the abovementioned works, in this letter, we focus on improving the lifespan of the IoT network and enabling a reliable link between the IoDs and the network with the help of UAV. Specifically, we consider the UAV to power energy-constrained IoDs that intend to transmit data to the network or base station (BS). However, the direct links between the IoDs and the BS are heavily obstructed, and therefore, the UAV provides decode-and-forward (DF)-based relay assistance to accomplish the communication. The proposed model can enable wireless communications in remote areas, emergency scenarios, disaster-affected regions, and sites where physical intervention for battery replacement or recharge is unrealistic. For the considered setup, we derive accurate expressions for the outage probability (OP), system throughput (ST), and energy efficiency (EE), considering HIs and composite fading channels.

Notations: The probability, probability density function (PDF), and cumulative distribution function (CDF) of a random variable X are denoted by $\Pr[\cdot]$, $f_X(\cdot)$, and $F_X(\cdot)$, respectively. $\Upsilon(\cdot,\cdot)$ and $\Gamma(\cdot,\cdot)$ are the lower and upper incomplete gamma functions, respectively, and $\mathcal{K}_\upsilon(\cdot)$ represents the modified Bessel function of the second kind of υth order [11, Eq. (8.407.1)].

II. SYSTEM AND CHANEL MODEL

The proposed system considers a UAV relay-assisted IoD network, as shown in Fig. 1, wherein energy-constrained multiple IoDs intend to transmit data to a remote BS. All the involved nodes communicate in half-duplex mode and are equipped with a single antenna. The

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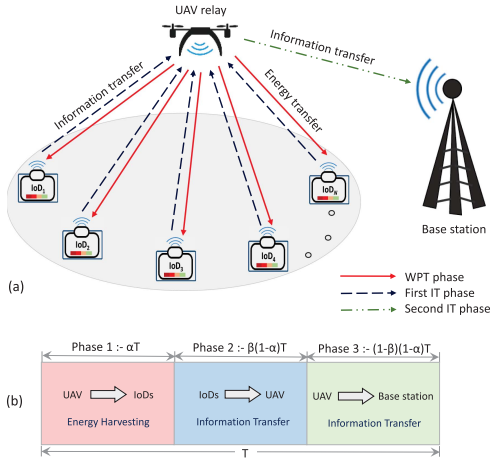

Fig. 1. (a) System model. (b) One transmission block.

distance between the IoDs and UAV is calculated as $d_{i,u} = \sqrt{h^2 + r_{i,u}^2}$, for $i \in \{1, \ldots, N\}$, where h is the UAV's height and $r_{i,u}$ is the horizontal distance between ith IoD and UAV. The elevation angles are given by $\theta_{i,u} = \arctan(h/r_{i,u})$. One transmission block of duration T is divided into three slots. In the first phase (αT duration), IoDs harvest energy from the RF signal transmitted by the UAV. Thereafter, one of the selected IoD transmits its signal to the UAV in the second phase ($\beta(1-\alpha)T$ duration). The best IoD-UAV link is selected by following the signal-to-noise-plus-distortion-ratios (SNDR)-based user selection technique. In the third phase ($(1-\beta)(1-\alpha)T$ duration), UAV applies DF operation to relay the signal to the BS.

We assume that the system operates in a block fading scenario, where the channel coefficient remains constant for the entire block duration of T. The channel coefficients between two communicating nodes $h_{i,j}$, with $(i, j) \in \{(i, u), (u, b)\}$ $\forall$ $i \in \{1, \ldots, N\}$, can be modeled in the form of a first-order autoregressive process using [12], taking into account the relative mobility between the nodes. As such, $h_{i,j} = \rho_{i,j}\hat{h}_{i,j} + \sqrt{(1-\rho_{i,j}^2)}h_{e_{i,j}}$, where $\hat{h}_{i,j}$ denotes the estimated channel gain and $h_{e_{i,j}} \sim \mathcal{CN}(0, \Omega_{e_{i,j}})$ represents the error component associated to mobility. $\rho_{i,j}$ is the correlation coefficient between $\hat{h}_{i,j}$ and $h_{i,j}$, and is expressed by using the Jakes's model as $\rho_{i,j} = \mathcal{J}_0(2\pi f_c v \tau_s/c)$, where $\mathcal{J}_0$ represents the first kind Bessel function of zeroth-order [11], c and f_c denotes the speed of light and carrier frequency, v represents the relative velocity between the communicating nodes, and τ_s is the symbol duration.

The composite channel model, i.e., Nakagami-m and log-normal distribution are used to characterize small-scale fading and shadowing of UAV links, whereas probabilistic line-of-sight (LoS) model is leveraged to compute the path-loss [13]. Average path-loss between the UAV-to-IoDs and UAV-to-BS link can be obtained as $\mu_{i,j} = (\mathcal{P}_{\text{LoS}}(\theta_{i,j})\mu_{\text{LoS}} + \mathcal{P}_{\text{NLoS}}(\theta_{i,j})\mu_{\text{NLoS}})$, where $\mathcal{P}_{\text{LoS}}$ and $\mathcal{P}_{\text{NLoS}}$ represent the probabilities of LoS and non-LoS (NLoS), respectively. Here, $\mathcal{P}_{\text{LoS}}(\theta_{i,j}) = 1/(1 + a\mathrm{e}^{-b(\theta_{i,j}-a)})$, and $\mathcal{P}_{\text{NLoS}}(\theta_{i,j}) = (1 - \mathcal{P}_{\text{LoS}}(\theta_{i,j}))$, where $\theta_{i,j}$ is the angle between the UAV and terrestrial nodes (expressed in degrees), a and b represent the environment-based parameters. Furthermore, μ_{LoS} and μ_{NLoS} denote mean path-loss for the LoS and NLoS components, respectively [10], expressed as

$$\mu_{\text{LoS}} = 20\log_{10}(4\pi f_c/c) + 20\log_{10}d_{i,j} + \psi_{\text{LoS}} \tag{1}$$

and

$$\mu_{\text{NLoS}} = 20\log_{10}(4\pi f_c/c) + 20\log_{10}d_{i,j} + \psi_{\text{NLoS}} \tag{2}$$

ψ_{LoS} and ψ_{NLoS} represent the propagation environment constants.

The performance of the considered system can be affected by imperfections in the hardware circuitry, which includes nonlinearity and imbalances in the amplifier and in-phase quadrature, respectively, and phase distortions [8]. Considering these transceiver HIs, the received signal can be represented as $y_{i,j} = h_{i,j}(x_i + \zeta_{i,j}) + n_{i,j}$, where $\zeta_{i,j} \sim \mathcal{CN}(0, k^2P_t)$ is the distortion noise due to transceiver impairments, P_t is the transmit power and $k = \sqrt{k_i^2 + k_j^2}$, where k_i and k_j are the factors that indicate the impairments levels of transmitter and receiver, respectively. Special case, i.e., $\zeta_{i,j} = 0$, refers to the conventional model with ideal hardware.

1) Signal-to-Noise-Plus-Distortion-Ratios (SNDRs): The energy harvested at the IoDs during the WPT phase, i.e., αT, is given by $\mathcal{E}_H = \eta P_u |h_{u,i}^{\text{WPT}}|^2 \alpha T$, where P_u is the transmit power of UAV and η is the energy conversion efficiency. Here, we forgo to model $h_{i,u}^{\text{WPT}}$ using the autoregressive process because in this phase, the signal is used for EH and additional noise just contributes to the amount of energy harvested. The power available at the IoDs is expressed as $P_i = \frac{\eta\alpha P_u |h_{u,i}^{\text{WPT}}|^2}{\beta(1-\alpha)}$. In the first information transmission (IT) phase, i.e., $\beta(1-\alpha)T$, the signal received at the UAV from the IoD in the presence of HIs is given as $y_{i,u} = (\sqrt{P_i}x_i + \zeta_{i,u})h_{i,u} + n_u$, where P_i and x_i refer to the transmit power and unit energy symbol transmitted by the IoD, respectively, $n_u \sim \mathcal{CN}(0, \sigma^2)$ represents the additive white Gaussian noise (AWGN) at UAV. There are multiple IoDs that intend to transmit their information to the BS via UAV, among which the best IoD is selected based on the instantaneous SNDR corresponding to the IoD-UAV link [7]. This can be mathematically given as

$$i^* = \arg \max_{i \in \{1,\ldots,N\}} \{\gamma_{i,u}\}. \tag{3}$$

Now, we incorporate the mobility model and substitute the value of P_i into the received signal from the selected IoD, given by $y_{i^*,u}$. Thus, the instantaneous SNDR at the UAV can be given as

$$\gamma_{i^*,u} = \frac{\chi\rho_{u,i^*}^2|\hat{h}_{i^*,u}|^2}{\chi k^2\rho_{u,i^*}^2|\hat{h}_{i^*,u}|^2 + \chi(1-\rho_{u,i^*}^2)(1+k^2)\Omega_{e_{i^*,u}} + \beta(1-\alpha)\sigma^2} \tag{4}$$

where $\chi = \eta\alpha P_u |h_{u,i^*}^{\text{WPT}}|^2$. Further, the UAV applies DF operation and transmits the signal to the BS. The signal received at the receiver is given by

$$y_{u,b} = (\rho_{u,b}\hat{h}_{u,b} + \sqrt{(1-\rho_{u,b})^2}h_{e_{u,b}})(\sqrt{P_u}x_u + \zeta_{u,b}) + n_b \tag{5}$$

where $n_b \sim \mathcal{CN}(0, \sigma^2)$ represents the AWGN at the receiver. Based on the received signal, the instantaneous SNDR at the BS can be given as

$$\gamma_{u,b} = \frac{\rho_{u,d}^2 P_u|\hat{h}_{u,b}|^2}{k^2\rho_{u,b}^2 P_u|\hat{h}_{u,b}|^2 + (1-\rho_{u,b}^2)P_u(1+k^2)\Omega_{e_{u,b}} + \sigma^2}. \tag{6}$$

III. PERFORMANCE ANALYSIS

1) Outage Probability (OP): We define the OP as an event that occurs when the instantaneous data rate falls below the predefined target rate. The OP is mathematically formulated as

$$\begin{aligned}\mathcal{P}_{\text{out}} &= 1 - (1 - \Pr[\mathcal{R}_{i^*,u} < r_{\text{th}}])(1 - \Pr[\mathcal{R}_{u,b} < r_{\text{th}}]) \\ &= 1 - (1 - \Pr[\gamma_{i^*,u} < \phi])(1 - \Pr[\gamma_{u,b} < \delta])\end{aligned} \tag{7}$$

where $\mathcal{P}_{i^*,u} = \Pr[\mathcal{R}_{i^*,u} < r_{\text{th}}]$ and $\mathcal{P}_{u,b} = \Pr[R_{u,b} < r_{\text{th}}]$ are termscorresponding to decoding of signal in the first and second IT phases $\mathcal{R}_{i^*,u} = \beta(1-\alpha)\log_2(1+\gamma_{i^*,u})$ and $\mathcal{R}_{u,b} = (1-\beta)(1-\alpha)\log_2(1+\gamma_{u,b})$, r_{th} is the target rate in terms of bps/Hz, $\phi = 2^{r_{\text{th}}/\beta(1-\alpha)} - 1$ and $\delta = 2^{r_{\text{th}}/(1-\beta)(1-\alpha)} - 1$ are the target SNRs for the IoD-to-UAV and UAV-to-BS links, respectively. Using [14, Eq.

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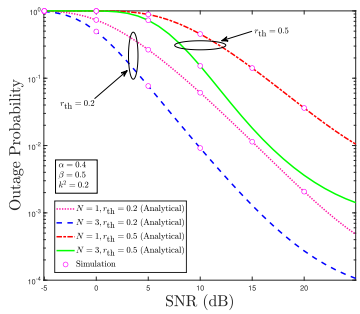

Fig. 2. OP versus SNR with different r_{th} and N.

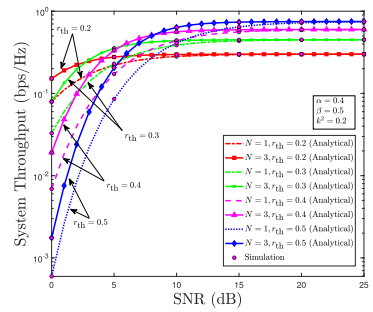

Fig. 3. ST versus SNR with different r_{th} and N.

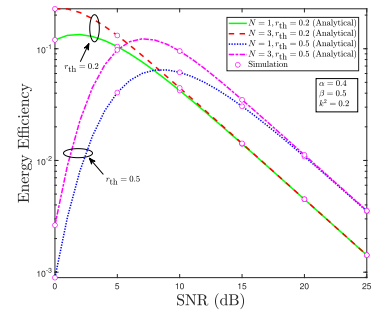

Fig. 4. EE versus SNR with different r_{th} and N.

(16)], (3), and (7), the expression for the OP can be obtained as

$$\mathcal{P}_{\text{out}} = 1 - \left(1 - \prod_{i=1}^{N} \Pr\left[\gamma_{i,u} < \phi\right]\right)(1 - \Pr[\gamma_{u,b} < \delta])$$
$$= 1 - (1 - (F_{\gamma_{i,u}}(\phi))^N)(1 - F_{\gamma_{u,b}}(\delta)). \quad (8)$$

Now, the expressions of $F_{\gamma_{i,u}}(\phi)$ is provided in Lemma 1.

Lemma 1: The expression of $F_{\gamma_{i,u}}(\phi)$ is expressed as

$$F_{\gamma_{i,u}}(\phi) = \sum_{j=0}^{J} \frac{w_j}{\sqrt{\pi}} \Upsilon\left[m_{i,u}, \frac{A_3 \phi m_{i,u}}{(A_1 - A_2\phi)\mathcal{A}_{i,u}}\right] + \sum_{j=0}^{J}\sum_{l=0}^{L} \frac{w_j w_l}{\Gamma[m_{i,u}]\pi}$$
$$\times \left[\Gamma\left(m_{i,u}, \frac{A_3\phi m_{i,u}}{(A_1 - A_2\phi)\mathcal{A}_{i,u}}\right) - \left(\frac{m_{i,u}}{\mathcal{A}_{i,u}}\right)^{m_{i,u}} \sum_{k=0}^{m_{u,i}-1} \frac{1}{k!}\right.$$
$$\times \left(\frac{A_4 \phi m_{u,i}}{\mathcal{A}_{u,i}}\right)^k \left(\frac{1}{(A_1 - A_2\phi)}\right)^{m_{i,u}} e^{-\left(\frac{A_3\phi m_{i,u}}{(A_1 - A_2\phi)\mathcal{A}_{i,u}}\right)}$$
$$\times \sum_{r=0}^{m_{i,u}-1} \binom{m_{i,u}-1}{r} \left(\frac{A_4(A_1 - A_2\phi)\phi m_{u,i}\mathcal{A}_{i,u}}{m_{i,u}\mathcal{A}_{u,i}}\right)^{m_{i,u}-r-k}$$
$$\left.\times\ 2(A_3\phi)^r K_{m_{i,u}-r-k}\left(2\sqrt{\frac{A_4\phi m_{i,u} m_{u,i}}{(A_1 - A_2\phi)\mathcal{A}_{u,i}\mathcal{A}_{i,u}}}\right)\right]. \quad (9)$$

Proof: Please see Appendix A.

On utilizing (6) and then performing some mathematical simplifications, we can express $F_{\gamma_{u,b}}(\delta) = \Pr[|\hat{h}_{u,b}|^2 < \Theta]$, where $\Theta = (\delta(1-\rho_{u,b}^2)P_u(1+k^2)\Omega_{e_{u,b}} + \delta\sigma^2)/(P_u\rho_{u,b}^2 - \delta k^2 \rho_{u,b}^2 P_u)$. Similar to (13), the pdf of $|\hat{h}_{u,b}|^2$ can be obtained and substituted in $F_{\gamma_{u,b}}(\delta) = \int_0^{\Theta} f_{|\hat{h}_{u,b}|^2}(\delta)dz$, to obtain the cdf $F_{\gamma_{u,b}}(\delta)$ by the aid of [11, Eq. (3.381.1)] as

$$F_{\gamma_{u,b}}(\delta) = \frac{1}{\sqrt{\pi}} \sum_{g=0}^{G} \frac{w_g}{\Gamma[m_{u,b}]} w_g \Upsilon\left[m_{u,b}, \Theta \frac{m_{u,b}}{\mathcal{A}_{u,b}}\right] \quad (10)$$

where $\mathcal{A}_{u,b} = 10^{(\sqrt{2}\delta_{u,b} t_g + \mu_{u,b})/10}$, w_g provides the weights of the Gauss–Hermite quadrature, and $[t_g]_{g=1}^{G}$ provides the roots of Hermite polynomial with G samples.

2) System Throughput (ST): Considering a delay-limited information transmission, ST is expressed as the average target rate that can be achieved over the fading channels. Therefore, the ST can be defined mathematically as

$$\mathcal{S}_T = \min\{\beta(1-\alpha)(1-\mathcal{P}_{i^*,u})r_{\text{th}}, (1-\beta)(1-\alpha)(1-\mathcal{P}_{u,b})r_{\text{th}}\}. \quad (11)$$

3) Energy Efficiency (EE): In an energy-constrained wireless network, it is desirable to achieve higher ST at minimum energy consumption [15]. Therefore, EE becomes essential for designing any efficient system model. EE for the considered network can be expressed as

$$\xi_{\text{EE}} = \frac{\min\{\beta(1-\alpha)(1-\mathcal{P}_{i^*,u})r_{\text{th}}, (1-\beta)(1-\alpha)(1-\mathcal{P}_{u,b})r_{\text{th}}\}}{(1+\alpha)P_u}. \quad (12)$$

IV. NUMERICAL RESULTS

Throughout this section, we set $N = \{1, 3\}$, $k^2 = 0.2$, the mobility error coefficients $\Omega_{e_{i^*,u}} = \Omega_{e_{u,b}} = 0.1$, and the average transmit SNR as P_u/σ^2. The coefficients for the Gauss–Hermite quadrature approximations are taken as $J = L = G = 15$ to achieve sufficient OP accuracy. The UAV's operational height, $h = 30$ m, and horizontal distance, $r_i = 50$ m, $f_c = 700$ MHz, $\tau_s = 10$ ms, and $v = 15$ km/h. The fading severity parameter is assumed to take integer values, and therefore, we set $m_{i,j} = 2$. Further, we consider a dense urban scenario for which the parameters are taken as follows: $a = 12.081$, $b = 0.1139$, $(c_1, d_1) = (9.64, 0.04)$, $(c_2, d_2) = (30.83, 0.04)$, $(\psi_{\text{LoS}}, \psi_{\text{NLoS}}) = (1, 20)$. The Monte-Carlo simulations are performed for 10^5 iterations to acquire accurate results.

The outage performance for different target rates with multiple IoDs is shown in Fig. 2. It is observed from the curves that OP at the BS increases with the increase in the target rate for the given set of parameter. This OP behavior is expected as the instantaneous rate that can be achieved when the fixed transmit SNR falls below the increased target rate value. Also, with the increase in the number of IoDs, the OP decreases significantly. This is because involving more IoDs in the network increases the diversity order and hence offers better OP performance.

Fig. 3 shows ST curves with varying transmit SNR and target rates. In a low transmit SNR region, the OP increases due to higher target rate, which results in lower ST. Moreover, with the increasing SNR value, the ST increases up to a maximum value, after which it attains saturation. This saturated value is considered as the maximum ST that can be obtained with that particular set of parameters. It is worth noting that the ST for comparatively high target rates saturates at higher SNR values. Another key observation is that with the increase in the number of IoDs, throughput in the low and mid-SNR regions is enhanced, however, no improvement can be obtained beyond the specific value of the achieved throughput.

Fig. 4 depicts the variation in energy-efficiency for different transmit SNR values and target rates. From the curves, it can be inferred that the system achieves higher EE in a particular range of transmit SNR. Initially, the harvested power is less in the low SNR region, so the EE is poor. Further, the EE at high SNR also degrades due to the saturated ST that is achieved after the mid-range of SNR values, as observed in

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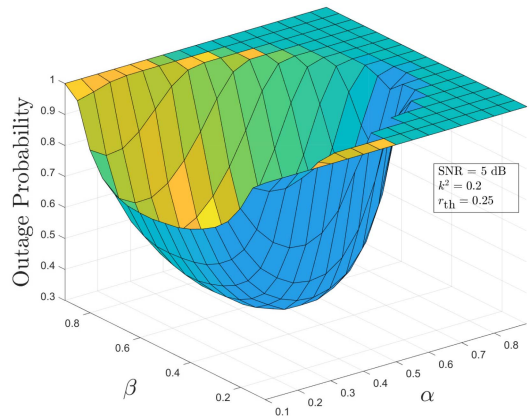

Fig. 5. OP versus α and β.

Fig. 3. It can be stated from Figs. 3 and 4 that the optimum performance in terms of system throughput and EE can be attained in the mid-range of the SNR values, i.e., 5 to 10 dB for higher target rate ($r_{\rm th} = 0.5$) and in the lower SNR range from -3 to 2 dB for the lower target rate, ($r_{\rm th} = 0.2$).

Fig. 5 shows the 3-D plot of OP performance with different values of α and β. In light of the fact that α and β control the time allocation for the WPT and overall communication between the IoD and BS in each phase, the combined impact of these parameters is of particular relevance. It can be observed from Fig. 5 that the OP plot follows the convex nature. Specifically, for the given set of parameters, the minimum OP is obtained at $\alpha = 0.32$ and $\beta = 0.41$.

V. CONCLUSION

This letter has addressed two essential challenges, i.e., improving the network's lifespan and establishing a reliable communication link between the IoDs and the BS via a UAV relay node. Practically, this model can be applied for various monitoring and data-gathering applications, e.g., precision agriculture where low-power IoDs can be exploited for crop monitoring and mapping to increase crop production. Here, we have investigated the performance of a WPT-enabled UAV-assisted IoT system in terms of OP, ST, and EE, considering the composite fading channel model, i.e., log-normal distribution for shadowing and the Nakagami-m distribution for small-scale fading. Simulation and analytical results have illuminated the impact of TS factor, target rate, and the number of IoDs in the cluster on the system performance.

APPENDIX A

Let $X = |h_{u,i}^{\rm WPT}|^2$ and $Y = |\hat{h}_{i,u}|^2$ denote the composite fading channels whose distributions are modeled as small-scale fading relatively varying with average power. The PDF of the composite fading channel can be obtained by leveraging the log-normal PDF for shadowing, i.e., $f_W(w)$ and PDF of gamma distribution for Nakagami-m faded signal amplitude distribution. Therefore, the PDF of X can be given as $f_X(x) = \int_0^\infty f_X(x/w) f_W(w) dx$, and obtained using the Gauss–Hermite quadrature technique [16], as

$$f_X(x) = \frac{1}{\sqrt{\pi}} \sum_{j=1}^{J} w_j \left(\frac{m_{u,i}}{\mathcal{A}_{u,i}}\right)^{m_{u,i}} \frac{x^{m_{u,i}-1}}{\Gamma[m_{u,i}]} e^{-\left(\frac{m_{u,i}x}{\mathcal{A}_{u,i}}\right)} \tag{13}$$

where $\mathcal{A}_{u,i} = 10^{(\sqrt{2}\delta_{u,i}t_j + \mu_{u,i})/10}$, w_j gives the weights of the Gauss–Hermite quadrature, and $[t_j]_{j=1}^{J}$ provides the roots of the Hermite polynomial with J samples. $\delta_{u,i} = c_1 {\rm e}^{-\theta_{u,i} d_1}$ is the standard deviation, wherein the environment parameters c_1 and d_1 varies for different scenarios. Similarly, the PDF of Y can be obtained as

$$f_Y(y) = \frac{1}{\sqrt{\pi}} \sum_{l=1}^{L} w_l \left(\frac{m_{i,u}}{\mathcal{A}_{i,u}}\right)^{m_{i,u}} \frac{x^{m_{i,u}-1}}{\Gamma[m_{i,u}]} e^{-\left(\frac{m_{i,u}y}{\mathcal{A}_{i,u}}\right)} \tag{14}$$

where $\mathcal{A}_{i,u} = 10^{(\sqrt{2}\delta_{i,u}t_j + \mu_{i,u})/10}$, $\delta_{i,u}$ is the standard deviation with environment parameters c_2 and d_2. By exploiting (4) and (7), the CDF of $F_{\gamma_{i,u}}(\phi)$ can be formulated as

$$F_{\gamma_{i,u}}(\phi) = \Pr\left[\frac{A_1XY}{A_2XY + A_3X + A_4} < \phi\right] = \int_0^{\frac{A_3\phi}{A_1 - A_2\phi}} f_Y(y) dy + \int_{\frac{A_3\phi}{A_1-A_2\phi}}^{\infty} f_Y(y) \int_0^{\frac{A_4\phi}{(A_1-A_2\phi)y - A_3\phi}} f_X(x) f_Y(y) dx dy \tag{15}$$

where $A_1 = \eta\alpha\rho_{i_u,u}^2 P_u$, $A_2 = \eta\alpha k^2 \rho_{i_u,u}^2 P_u$, $A_3 = \eta\alpha(1-\rho_{i_u,u}^2)P_u(1+k^2)\Omega_{e_{i_u,u}}$, and $A_4 = \beta(1-\alpha)\sigma^2$. Now, inserting (13) and (14) in (15) followed by simplifying the obtained integral expression using [11, Eqs. (3.381.1), (3.381.6), and (8.352)], we can get $F_{\gamma_{i,u}}(\phi)$, as in (9).■

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